

Fluorine

F (F I). Strongest classified transitions below 2 μm (NIST): 27 levels, 46 lines.

F⁺ ionisation limit 140524.50 cm⁻¹ = 17.42282 eV (71.162 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

